AutoContDamage: Automated Container Damage Detection in Ports Using RF-DETR Model for Enhanced Operational Efficiency

***Abstract*—Container damage has become a ubiquitous threat as the exponential growth of world trade requires efficient logistics and transportation systems. Despite progress in detecting container damage, finding subtle damage is still challenging due to limited data and complex visual conditions. This paper proposes a method which uses RF-DETR to detect complex and subtle issues in container structures, while leveraging the data set with data augmentation to improve the detection of small defects such as holes, dents, rusts, etc.This research is among the first to apply RF-DETR to container damage detection, addressing the complex conditions of seaports and analyzing the performance of deep learning models in terms of accuracy and robustness, under real-world conditions. This method shows that RF-DETR achieved a mean average precision (mAP) of 94.1%, a precision rate of 92.6%, and a recall of 90.0%. Ultimately, the research contributes toward reducing inspection delays, minimizing human error, and improving operational efficiency in containerized logistics.**

***Index Terms*—Container damage, Object detection, Data Augmentation, Deep learning, RF-DETR.**

I. INTRODUCTION

Containerization is crucial in international shipping sector, handling over 50% of the total value of seaborne trade by container ships worldwide[1]. Shipping containers are a key element of global logistics and enable efficient transport of goods over long distances [2]. Unfortunately, containers are prone to damage such as structural deformations, corrosion, corner castings, floor damage, rust, and infestations, etc. Currently, customs officers check the surface of shipping containers during the transition at the port terminal to detect damages manually that cause delays and poor predictability. Apparently, manual inspection methods are time-consuming, labor-intensive, and possibly hazardous[3].During loading, unloading, and transhipment procedures, container damage occurs in port environments and leads to significant challenge that adversely affects operational efficiency, equipment integrity, and supply chain accountability[4].To prevent the consequences of damaged containers, a timely inspection and detection is necessary. In general, the rapid growth and increasing complexity of global container shipping highlight the need for advanced solutions in port operations. Figure 2 demonstrates some container damage that is crucial to detect damages and further ensure the implementation of an effective automated detection system.

Recent research has explored container damage detection systems, which primarily use deep learning approaches including YOLO models and CNNs for container handling, focuses on automation in container recognition[5][6][7]. However, these systems face significant limitations in port environments including poor performance in varying lighting conditions, limited real-time processing capabilities, high computational requirements, and challenges with subtle damages such as small dents, cracks and rusts remains challenging[8][9]. In this area, applying advanced lightweight detection models marks a notable breakthrough. To identify and classify multiple types of container damages from real-world port images, the proposed study intends to develop a deep learning-based approach using RF-DETR. By leveraging the capabilities of RF-DETR (transformer-based), the research aims to improve accuracy for container logistics.

The key contributions of this research are highlighted below:

1) The proposed model implemented data augmentation strategies, including brightness and geometric transformations, thereby improving robustness against real-world environmental conditions.
2) Determined how RF-DETR performs better in accurately detecting subtle and small-scale container damages, particularly dents, holes, and rusts with a mean average precision (mAP) of 94.1%, precision rate of 92.6%, and a recall of 90.0% .
3) The study validates the integration of container damage detection, aiming to reduce manual errors, and enhance operational safety.

The study comprises five sections: Section II reviews container damage detection studies, Section III details the proposed system, Section IV presents results and comparisons, and Section V concludes with future directions.

II. RELATED WORKS

Recent advances in deep learning have significantly improved road damage detection. We summarize key contributions in Table I. Below, we review these works thematically.

Nguyen Thi Phuong et al. [8] proposed a YOLO-NAS based approach for container inspection, and they demonstrated good performance of YOLO-NAS with a precision of 92.4%, recall of 84.1%, and mAP of 91.2%. Their results demonstrated the model's ability to detect not just subtle and complex damage in real time but also its potential to facilitate next-generation port operations. However, the authors noted challenges with regards to detecting very small defects as well as computational demands, and dataset reliance on well-annotated datasets. Kumar [9] compared YOLOv11, YOLOv12, and RF-DETR

TABLE I: Summary of Related Works

Ref.	Year	Dataset	Method	Result	Limitations
[10]	2020	Custom Container Dataset	DB-YOLOv8 with CA attention	mAP 83.4%, detection speed 90.1 FPS	Only surface defects; ignores mislabeling, structural deformation, hidden damage; dataset generalization limited
[11]	2021	Experimental port handling setup	Impact Detection Methodology (IoT acceleration sensors)	No accuracy metrics reported	Single-axis data; noisy in chaotic conditions; cannot detect visible damage, mislabeling, structural issues
[12]	2024	Custom container dataset	Multi-type damage detection with MobileNetV2 and Transfer Learning	No standard accuracy, precision, or recall metrics	Dataset imbalance; small sample size; inconsistent detection of rare damages; ignores hidden/mislabeled/structural defects
[13]	2025	Custom video dataset	Dual-stage YOLO (door localization + seal detection) with dehazing spatiotemporal tracking	mAP50 0.93, Precision 0.90, Recall 0.85	Limited data diversity; performance drops at high speed (mAP50 0.68); only seal defects
[14]	2025	Custom container dataset	DeepLabv3+ (semantic segmentation)	Corrosion detection 49% accuracy, overall 76%	Low accuracy; small dataset; focused only on corrosion; ignores other damages
[9]	2025	Roboflow Container Dataset	YOLOv11, YOLOv12, RF-DETR	YOLOv11 – mAP@50: 81.9%, Precision: 91.9% YOLOv12 – mAP@50: 81.9%, Precision: 78.9% RF-DETR – mAP@50: 77.7%, Precision: 83.8%	Small dataset ,limited focus on subtle or rare damages, only visible structural damage
[8]	2025 -	Roboflow Container Damage Dataset	YOLO-NAS	Precision 92.4%, Recall 84.1%, mAP 91.2%	Difficulty detecting very small defects

using 278 annotated images for all three models. They reported mixed results, indicating that YOLOv11 and YOLOv12 both had a mAP score of 81.9%; RF-DETR had a mAP score of 77.7% but demonstrated better localization capability in more complex examples, such as fire-damaged containers or bent containers. These authors stated in their conclusion paper that their study demonstrated the success of YOLOv11, YOLOv12 and RF-DETR for large-scale damage annotations; however, the dataset size was small, which limited coverage for granularity in subtle damages and anomalies, as they did not extend beyond visible structural damage.

In another approach, Ornelas et al. [14] attempted to identify corrosion in cargo containers using a semantic segmentation approach using a DeepLabv3+ model with a base model and with a fine-tuned model tested against a custom dataset describing 500 annotated images. For detecting corrosion (examples only), their fine-tuned model demonstrated a 49% accuracy, and 76% accuracy including background, under deepLabv3+, as a whole. While the authors considered their study as a pilot protocol study, we see the strength of their study being their ability to achieve pixel-level precision and localization for corrosion, and also, in establishing a custom dataset made that breaks new ground in ORM in this study space of cargo container maintenance status.

Wang et al. [12] proposed a framework for identifying damage to containers that leverages MobileNetV2 and transfer learning to categorise nine different categories of visual damage. After more than 5,000 cycles of training and validation in actual port scenarios, the model exhibited great accuracy for typical damage types while keeping computing economy because of its lightweight design. Its efficiency and capacity to manage numerous damage kinds are its key advantages; on the other hand, its limitations include an imbalanced dataset, a short sample size, and inconsistent identification of less prevalent problems. Additionally, the study only looks at surface defects that are immediately evident; it misses problems like mislabeling, structural deformities, or concealed anomalies.

To increase recognition when defects are highly fused with the container backdrop or have little color differences. Liu et al. [10] introduced the DB-YOLOv8 method for container defect detection. This algorithm extends the YOLOv8 backbone with a Coordinate Attention (CA) mechanism. The model met the requirements for real-time inspection with a mean average precision (mAP) of 83.4%, which was 2.4% higher than the baseline YOLOv8. It also displayed a detection speed of 90.1 FPS. This study's key strength is its capacity to decrease background noise, concentrate on aspects that are related to flaws, and obtain high accuracy and efficiency. Nevertheless, there are drawbacks, such as the assessment of surface flaws only, ignoring other anomaly types like mislabeling, structural deformation, or hidden damage, and possible difficulties extrapolating results outside of the tested dataset.

Yu [13] used a two-stage YOLO framework to create an automated broken seal detection system for containers that is based on video. Container doors are located in the first stage, and seal faults are found in the second stage utilizing spatiotemporal tracking and dark channel dehazing. The model beat single-stage models and conventional approaches, reaching mAP50 of 0.93, precision 0.90, and recall 0.85. Real-time detection and resilience to changing illumination and occlusion are among its advantages. Restrictions include reliance on GPU hardware, low performance at high container speeds (mAP50 0.68), and limited data diversity. Although it is focused on seal detection, it does not address structural concerns, mislabeling, or general container deterioration; rather, it offers a solid framework for video-based monitoring.

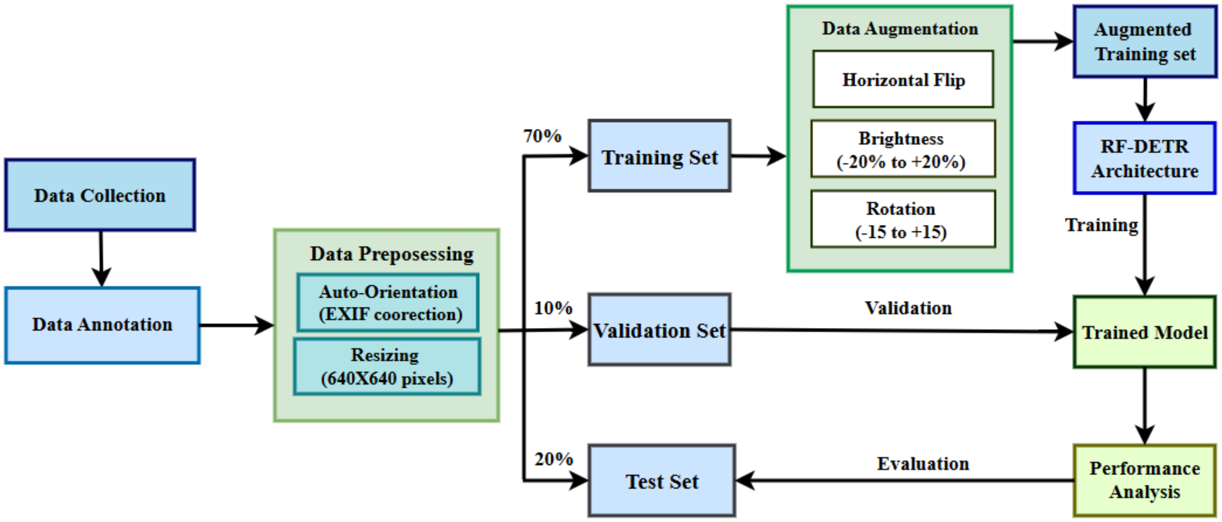


Fig. 1: The Workflow of the Proposed RF-DETR Model.

Fig. 2: The Sample of Annotated Container Damage Dataset.

Using IoT acceleration sensors, Jakovlev et al. [11] developed an Impact Detection Methodology (IDM) to track physical impacts on containers while they are being handled. The approach promotes operational safety and predictive maintenance by efficiently collecting major impacts, particularly along the vertical axis. Identifying otherwise invisible effects and giving data-driven insights are among its advantages. Its limitations include reliance on single-axis data, noise from environmental factors, inconsistent detection in chaotic conditions, and failure to address structural issues, mislabeling, or visible damage.

Based on the above literature review, we found that current research is based on small or imbalanced data, focuses on limited types of container damage, and has moderate accuracy that requires improvement. These limitations highlight the need for a more comprehensive approach, so we propose this research using a larger dataset and a RF-DETR model to improve detection performance.

III. Methodology

Figure 1 shows a block diagram for the suggested container damage detection system. The container damage detection system is made up of four parts: i) data collection, ii) dataset annotation and preprocessing, iii) model training with RF-DETR, and iv) model evaluation and performance analysis.

A. Dataset

The dataset for this study was collected from the open-source repository Roboflow. It comprises 5,761 images of shipping containers with three annotated damage classes: dent, hole, and rust. The dataset was exported in COCO format for compatibility with deep learning frameworks.

B. Data Preprocessing

To ensure consistency and standardization across the dataset, preprocessing steps were applied prior to training. These included:

1) Auto-Orient: Correcting image orientation using embedded EXIF metadata.
2) Resizing: All images were resized to 640 × 640 pixels, ensuring uniformity of input dimensions.

These preprocessing steps reduce variations in image orientation and scale, thereby enhancing the quality of feature extraction.

C. Augmentation

The dataset was divided into three subsets: 70% for training, 20% for validation, and 10% for testing. To enhance dataset variability and reduce overfitting, data augmentation was applied only to the training set. The augmentation strategies included:

1) Horizontal flipping of images.
2) Brightness adjustment within the range of -20% to +20%.
3) Rotation between -15° and +15°.

Following augmentation, the dataset size increased from 5,761 images to 9,652 images, out of which 6740 images were used for training, 1929 for validation, and 983 for testing.

D. Proposed RF-DETR Architecture

The system uses the RF-DETR (RoboFlow Detection Transformer) model for container damage detection. RF-DETR, released in March 2025, builds upon the original DETR framework by Facebook AI Research (2020) and integrates improvements from Deformable DETR and LW-DETR while leveraging the DINOv2 backbone.[9] Unlike traditional CNN-based object detectors, RF-DETR employs a transformer encoder-decoder architecture to process image features and predict object locations and classes, eliminating the need for region proposals, anchor boxes, and complex components like non-maximum suppression (NMS). This results in a cleaner and more unified architecture. The key characteristics of RFDETR include:

1) Deformable attention modules to capture features across multiple scales, which is critical for detecting small or irregular container damages.
2) Transformer backbone with DINOv2 for efficient feature aggregation and enhanced background modeling, improving performance in complex container environments.
3) High accuracy with relatively efficient computation, demonstrating a mAP of 60.5 at 25 FPS on the MS COCO dataset.

This makes RF-DETR particularly suitable for detecting damage types with varying shapes and textures in complex container environments.

IV. EXPERIMENTAL ANALYSIS

The experimental setup used for training and evaluating the proposed model RF-DETR, the evaluation metrics applied, and the results obtained from the experiment.

A. Experimental Setup

The Small RF-DETR model is used to detect container damages such as dents, holes, and rusts. To optimize the training and fine-tuning process,a powerful NVIDIA A100-SXM4 GPU with 80GB VRAM was utilized, supported by the Google Colab Pro environment. The experimental setup was based on Python 3.12 runtime with CUDA 12.4, ensuring efficient training, checkpoint management, and model evaluation. The software programs that were used: rfdetr = = 1.2.1, supervision = = 0.26.1 and roboflow. Overll, the batch size 16 and gradient accumulation steps of 1 were choosen TO balance processing speed and memory utilization.

TABLE II: Per-Class Test Evaluation Metrics of RF-DETR Model

Model	mAP[50:95]	mAP[50]	Precision	Recall
Dent	0.684	0.952	0.984	0.84
Rust	0.635	0.935	0.972	0.84
Hole	0.586	0.857	0.911	0.84
all	**0.635**	**0.914**	**0.955**	**0.84**

TABLE III: Comparing Proposed RF-DETR Model performance with other models

Model	Accuracy	Precision	Recall
YOLO-NAS	0.912	0.924	0.8675
YOLOv8	0.636	0.851	0.72
Yolov11	0.819	0.919	0.761
Yolov12	0.819	0.789	0.784
RF-DETR (Proposed)	**0.94**	**0.92**	**0.90**

B. Results

To demonstrate the effectiveness of the trained model, the study projected on-road images from the test dataset. All tests were performed using the same environments and parameters. Figure 3 demonstrates the training and validation curves across 55 epochs. The training versus validation loss curve shows a consistent downward trend, where the best validation loss recorded was 4.515, demonstrating effective generalization without severe overfitting. The training versus validation accuracy curve indicates that the validation accuracy stabilized around 99.51%, confirming strong convergence of the model. Further, figure 4 presents the performance metrics across epochs. In figure 4a, the precision curve indicates a steady increase throughout training and stabilizes at 0.94 confirming the model's effectiveness in reducing false positives. figure 4b indicates that the recall curve is continuously improving as the maximum point is 0.90, illustrating the fact that the model can capture major true damaged containers. In figure 4c, the mAP curve rises rapidly in the early epochs and plateaus at 0.93, confirming consistent detection accuracy across damage categories. Given Table II is showing the difference among the three classes three classes: dent, rust, and hole in terms of precision, recall, mAP50, mAP50-95, demonstrating accurate classification.Figure 5 shows the observable outcomes of these assessments using the proposed RF-DETR model, displaying accurate detections of damaged containers across various scenarios.Moreover, testing the proposed RF-DETR model with different configurations and achieved superior results with an overall accuracy of 0.94, precision of 0.92, and recall of 0.90. this model outperformed YOLO-NAS, YOLOv8, YOLOv11, and YOLOv12.The comparative analysis result is summarized in the given Table III.

V. Conclusion and Future Works

In this research, we propose an automated container damage detection system utilizing the RF-DETR model, demonstrating high efficacy in identifying complex and subtle damages such as dents, holes, and rust. The model achieved superior

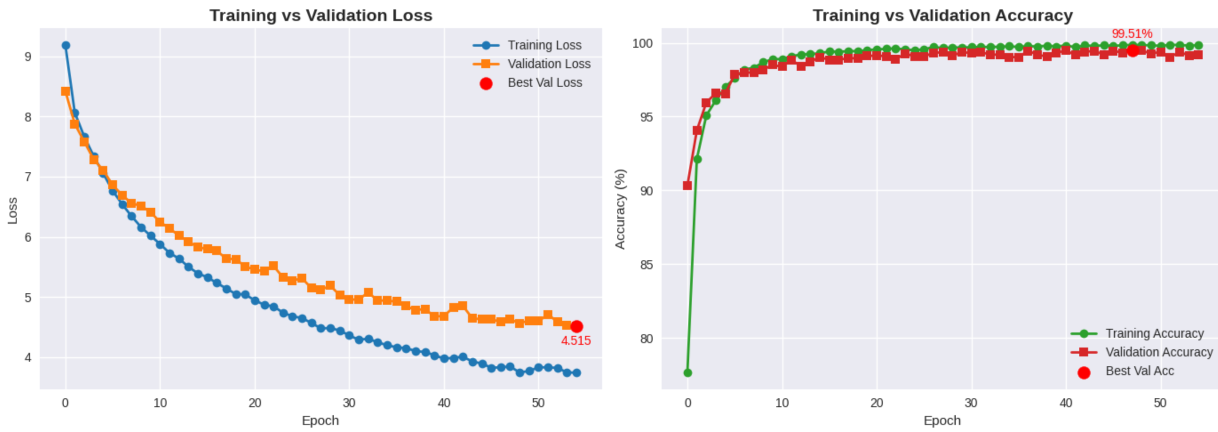
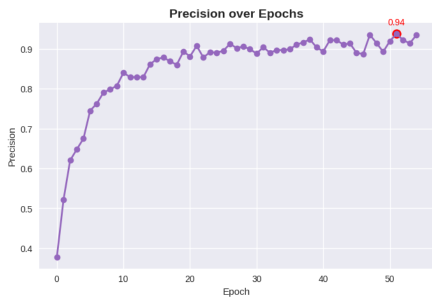
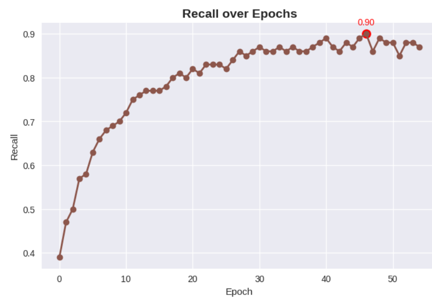


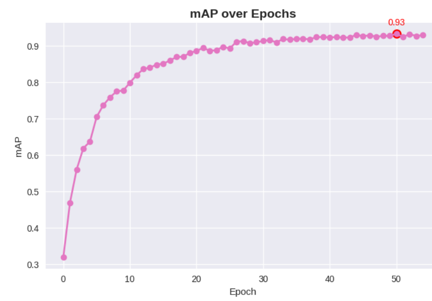
Fig. 3: Training vs Validation Loss and Accuracy



(a) Precision

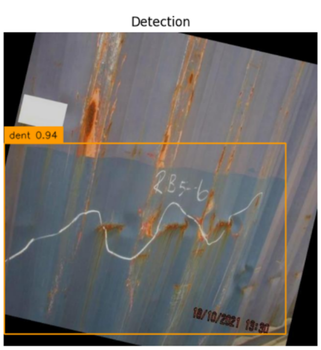


(b) Recall

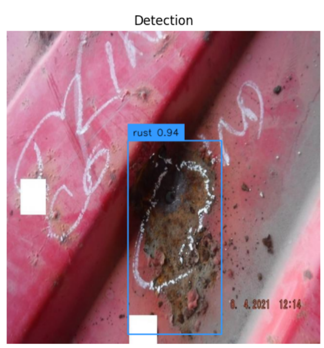


(c) mAP

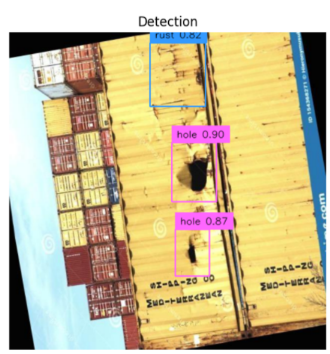
Fig. 4: The proposed model performance metrics across epochs.



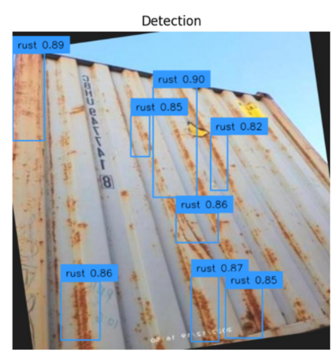
(a) Dent



(b) Rust



(c) Hole



(d) Multiple rust

Fig. 5: Container Damage Detections using RF-DETR.

performance with a mean Average Precision (mAP) of 94.1%, a precision of 92.7%, and a recall of 90.0%, outperforming existing YOLO-based benchmarks. This approach significantly enhances inspection reliability and speed, offering a robust solution to reduce manual inspection delays, minimize human error, and improve operational safety within port logistics. The model's performance is sensitive to hyperparameter configurations and requires substantial computational resources for training which is the limitation of our study.

Future research will focus on overcoming the model's sensitivity to hyperparameters and its high computational demands through automated optimization techniques and model compression. Further efforts will also explore scaling the model to a larger variant for increased accuracy, optimizing it for real-time inference speeds suitable for dynamic port environments.

References

[1] D. Song, "A literature review, container shipping supply chain: Planning problems and research opportunities," *Logistics*, vol. 5, no. 2, p. 41, 2021.

[2] A. Grzelakowski, “Global container shipping market development and its impact on mega logistics system,” *TransNav: International Journal on Marine Navigation and Safety of Sea Transportation*, vol. 13, no. 3, pp. 529–535, 2019.

[3] Z. Bahrami, R. Zhang, T. Wang, and Z. Liu, “An end-to-end framework for shipping container corrosion defect inspection,” *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1–14, 2022.

[4] S. Jakovlev, T. Eglynas, V. Jankunas, M. Jusis, and M. Voznak, “Analysis of damage to shipping container sides during port handling operations,” *Journal of Marine Science and Engineering*, vol. 13, no. 5, p. 982, 2025.

[5] H. Lee, I. Chatterjee, and G. Cho, “Ai-powered intelligent seaport mobility: enhancing container drayage efficiency through computer vision and deep learning,” *Applied Sciences*, vol. 13, no. 22, p. 12214, 2023.

[6] S. Bandong, Y. Y. Nazaruddin, and E. Joelianto, “Container detection system using cnn based object detectors,” in *2021 International Conference on Instrumentation, Control, and Automation (ICA)*. IEEE, 2021, pp. 106–111.

[7] H. Li, Y. Yang, S. Wang, Z. Chen, and L. He, “Automatic detection and extraction of lost shipping containers based on yolo and the segment anything model,” *Remote Sensing Letters*, vol. 15, no. 10, pp. 1023–1034, 2024.

[8] T. Nguyen Thi Phuong, G. S. Cho, and I. Chatterjee, “Automating container damage detection with the yolo-nas deep learning model,” *Science Progress*, vol. 108, no. 1, p. 00368504251314084, 2025.

[9] S. Kumar, “Container damage detection using advanced computer vision model yolov12 vs yolov11 vs rf-detr a comparative analysis,” *arXiv preprint arXiv:2506.22517*, 2025.

[10] Y. Liu, F. Jia, L. Yu, X. Li, Y. Liu, G. Chen, G. Wu, Y. Yin *et al.*, “Research on container defect detection based on db-yolov8,” *Academic Journal of Computing & Information Science*, vol. 7, no. 11, pp. 135–141, 2024.

[11] S. Jakovlev, T. Eglynas, M. Voznak, M. Jusis, P. Partila, J. Tovarek, and V. Jankunas, “Detection of physical impacts of shipping containers during handling operations using the impact detection methodology,” *Journal of Marine Science and Engineering*, vol. 10, no. 9, p. 1250, 2022.

[12] Z. Wang, J. Gao, Q. Zeng, and Y. Sun, “Multitype damage detection of container using cnn based on transfer learning,” *Mathematical Problems in Engineering*, vol. 2021, no. 1, p. 5395494, 2021.

[13] S. Yu, “Detection of broken seals on containers from video recordings based on yolo model.”

[14] D. Ornelas, D. Canedo, and A. J. Neves, “Semantic segmentation of corrosion in cargo containers using deep learning,” *Sustainability*, vol. 17, no. 14, p. 6480, 2025.